

RECIFE, Brazil

WMO: 828990

Lat: 8.126S

Lon: 34.924W

Elev: 33

StdP: 14.68

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	70.7	71.5	64.0	89.3	77.1	64.5	90.9	77.1	22.9	80.1	21.3	79.7	5.6	230	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.6	93.1	80.8	91.7	79.9	91.0	79.5	81.6	90.3	80.8	89.2	80.1	88.2	12.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.3	87.2	78.6	148.8	86.7	77.4	142.8	85.1	45.4	90.5	44.5	90.1	43.8	88.5	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	18.0	16.5	68.7	92.8	2.0	3.8	67.2	95.5	66.1	97.7	65.0	99.9	63.5	102.7		
			65.7	82.2	2.3	2.5	64.0	84.0	62.7	85.4	61.4	86.8	59.7	88.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.3	83.1	83.8	83.5	82.3	81.0	79.2	78.2	78.2	79.8	81.5	82.5	83.1
	DBStd	2.94	2.42	2.50	2.51	2.36	2.17	2.08	2.03	2.06	2.12	2.11	1.93	1.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11440	1027	946	1039	970	961	875	874	876	893	978	975	1026
	CDD65	5965	562	526	574	520	496	425	409	411	443	513	525	561
	CDH74	62748	6611	6429	7077	5976	5097	3585	3051	3246	4022	5327	5839	6488
	CDH80	19994	2435	2593	2905	2212	1555	807	591	608	833	1419	1804	2232
(14) Wind	WSAvg	9.5	9.3	9.2	8.4	8.1	8.2	9.0	10.1	10.7	11.1	10.6	10.0	9.7
(15) Precipitation	PrecAvg	90.7	4.1	5.4	9.3	11.7	12.8	14.8	13.3	7.8	4.4	2.1	1.7	2.6
	PrecMax	151.9	10.8	14.8	24.6	30.4	29.8	27.9	26.6	24.6	13.2	5.9	8.7	7.9
	PrecMin	57.2	0.0	0.7	1.1	2.2	2.4	4.9	2.5	1.9	0.8	0.0	0.1	0.3
	PrecStd	19.4	2.7	3.9	5.3	6.5	5.2	5.9	5.7	4.1	2.9	1.6	1.4	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	93.9	94.7	92.8	91.6	89.1	88.0	87.9	89.5	91.6	92.5	93.3
		MCWB	80.6	81.2	81.7	81.7	80.2	78.7	77.7	77.5	77.9	78.8	79.8	80.4
	2%	DB	91.6	92.8	92.9	91.2	89.7	87.5	86.2	86.2	87.6	89.8	91.1	91.4
		MCWB	79.6	80.5	81.0	80.4	79.6	78.2	76.9	76.4	77.1	78.2	79.0	79.5
	5%	DB	89.8	91.2	91.0	89.5	87.8	85.8	84.6	84.9	85.9	87.9	88.2	89.4
		MCWB	78.7	79.5	80.0	79.5	79.0	77.0	76.1	75.7	76.0	77.4	77.4	78.3
	10%	DB	87.9	89.1	88.2	87.7	86.1	84.2	82.9	83.0	84.2	85.8	86.2	87.0
		MCWB	77.8	78.6	78.7	78.7	77.8	76.0	75.0	74.6	74.9	76.0	76.1	77.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.3	82.5	83.0	82.5	81.5	79.9	79.3	78.9	79.2	80.0	80.6	81.2
		MCDB	90.7	91.9	92.3	90.7	88.8	86.6	85.5	85.8	86.8	89.3	90.9	91.1
	2%	WB	80.2	81.0	81.4	81.0	80.3	78.9	77.9	77.4	77.8	78.9	79.4	80.0
		MCDB	89.7	90.2	90.1	88.5	87.2	85.2	84.0	84.2	85.3	88.5	89.2	89.5
	5%	WB	79.4	80.0	80.5	80.1	79.5	78.0	77.0	76.5	76.8	77.8	78.2	79.0
		MCDB	88.1	88.7	88.7	87.1	86.0	83.8	83.0	83.0	83.9	86.4	87.1	87.3
	10%	WB	78.6	79.2	79.7	79.4	78.7	77.1	76.1	75.6	75.8	76.7	77.2	78.0
		MCDB	86.6	87.3	87.2	86.1	84.7	82.4	81.7	81.6	82.4	84.3	85.1	85.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	10.2	10.5	10.4	10.2	10.3	10.2	9.0	8.6	8.7	8.8
		MCDBR	11.6	12.1	12.6	12.4	12.4	11.9	11.9	12.0	11.0	11.8	11.4	12.0
	5% WB	MCWBR	4.7	4.9	5.0	5.3	5.4	5.3	5.5	5.6	5.0	5.0	4.8	5.0
		MCWBR	10.4	11.0	11.3	11.4	11.3	11.0	11.2	11.1	10.1	10.4	10.5	10.5
(40) Clear-Sky Solar Irradiance	taub		0.394	0.398	0.397	0.393	0.386	0.390	0.391	0.395	0.425	0.421	0.399	0.404
		taud	2.503	2.497	2.507	2.538	2.559	2.534	2.517	2.471	2.361	2.377	2.451	2.455
	Ebn at Noon		300	298	293	285	279	272	275	281	281	287	296	296
		Edh at Noon	36	36	36	33	31	31	32	35	40	41	38	38
(44) All-Sky Solar Radiation	RadAvg		1963	1963	1930	1749	1544	1399	1474	1649	1859	1982	2101	2031
	RadStd		126	96	101	122	122	92	103	77	95	94	90	87

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		-1.42	-0.81	-1.42	-3.11	-1.17	-0.67	N/A	N/A	-596
(47) Regional (3 neighbors)		-1.15	-0.94	-1.67	-1.42	-1.19	-1.08	N/A	N/A	-475

Nomenclature: See separate page